

Volumes of Revolution

ANSWER KEY



The disk and washer methods

- 1 Find the volume generated when the region under $y = \sqrt{x}$ on $[0, 4]$ is revolved about the x -axis.
$$V = \pi \int_0^4 (\sqrt{x})^2 dx = \pi \int_0^4 x dx = \pi \left[\frac{x^2}{2} \right]_0^4 = 8\pi.$$
- 2 The region between $y = x^2$ and $y = 4$ is revolved about the x -axis. Using the washer method, set up and evaluate the volume integral.
Bounds $x = -2$ to 2 :
$$V = \pi \int_{-2}^2 [4^2 - (x^2)^2] dx = \pi \int_{-2}^2 (16 - x^4) dx = \pi \left[16x - \frac{x^5}{5} \right]_{-2}^2 = \frac{384\pi}{5}.$$

The shell method

- 3 The region under $y = x^2$ on $[0, 2]$ is revolved about the y -axis. Use the shell method to find the volume.
$$V = 2\pi \int_0^2 x \cdot x^2 dx = 2\pi \int_0^2 x^3 dx = 2\pi \left[\frac{x^4}{4} \right]_0^2 = 8\pi.$$
- 4 The region bounded by $y = x$, $y = 0$, and $x = 3$ is revolved about the y -axis. Find the volume using the shell method.
$$V = 2\pi \int_0^3 x \cdot x dx = 2\pi \left[\frac{x^3}{3} \right]_0^3 = 18\pi.$$